



CERTIFICATE OF TESTS

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Printed: 03/24/2025

Heat: 24175231
Material: 5L-X52 4-1/2X.337 (15.00#)BR,DRL,PEB (11000102)
Vendor: Big River Steel - Osceola AR US (180431)

Purchase Order: 26708947
Sales Order: 1001001265
MFG Date: 01/30/2025

CHEMICAL PROPERTIES

	C	MN	P	S	NI	CR	CU	MO	SI	TI	V	CE
Ladle	0.0500	1.0400	0.0110	0.0010	0.0500	0.0500	0.1000	0.0200	0.1900	0.0010	0.0030	0.1202
	Al: 0.0260		Sn: 0.0050		N: 0.0090		B: 0.0004		Ca: 0.0022		Nb: 0.0020	
Chem	0.0617	1.1407	0.0101		0.0491	0.0557	0.1017	0.0145	0.1953	0.0008	0.0039	0.1365
	Al: 0.0294		Sn: 0.0048		N: 0.0135		B: 0.0002		Ca: 0.0027		Nb: 0.0053	
Chem	0.0689	1.1411	0.0105		0.0492	0.0555	0.1030	0.0148	0.1934	0.0013	0.0037	0.1438
	Al: 0.0285		Sn: 0.0048		N: 0.0152		B: 0.0003		Ca: 0.0026		Nb: 0.0048	

Acceptance Criteria: CEPCM = 0.2500

MECHANICAL PROPERTIES

YS (ksi): 70.10 TS (ksi): 74.80 EL (%) : 30.00 HRB: 86.00 Y/T(%): 0.94

CHARPY TEST RESULTS (BODY)

Orientation : TRANSVERSE	Specimen(S)	Energy (ft-lb)	%Shear
Spec. Size (in) : .394"X.197"X2.165"	1	78.00	100.00
Test Temp (°F) : -20	2	82.00	100.00
Location : 3T90	3	79.00	100.00
	Average	79.67	100.00

Acceptance Criteria Min. (ft-lb) : 10

COMMENTS

MINIMUM HYDROSTATIC TEST PRESSURE (PSI) : 4,700 PER 5 SECONDS

MANUFACTURED TO THE 46TH EDITION OF API 5L SPECIFICATION
MANUFACTURED IN THE USA-HIGH FREQUENCY WELDED PRODUCT
WELD SEAM ANNEALING AT 1600 DEGREES F.MINIMUM
ULTRASONIC TESTED TO N-10 NOTCH 1/8 DRILLED HOLE
FLATTENING TEST PASSED
1 INCH L90 STRIP SPECIMEN
THIS MATERIAL MEETS :ASTM A53B-24,ASME SA53B-21 LATEST EDITION
API 5L PSL 2 GRADE X52M
MEETS HARDNESS REQUIREMENTS OF NACE MRO 175/ISO 15156-2, CLAUSE A.2.1.1
MEETS ISO 10474 3.1.B/EN 10204: 3.1 / STANDARDS ARE LATEST REVISION
MELTED AND MANUFACTURED IN THE U.S.A.
STEEL SUPPLIER: Big River Steel - Osceola AR US
COIL MANUFACTURER: Big River Steel - Osceola AR US

The above analyses and tensile properties are correct to the best of my knowledge and belief.

Quality Control Dept